

DS	Session	Graph	Question Information	Level 1 Challenge 82
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Question description

Udaya family is going to order a large pizza and eat it together. A total of n family members will join the order, and there are m possible toppings. The pizza may have any number of toppings.

Each family member gives two wishes concerning the toppings of the pizza. The wishes are of the form "topping x is good/bad". Your task is to choose the toppings so that at least one wish from everybody becomes true (a good topping is included in the pizza or a bad topping is not included).

Input

The first input line has two integers n and m : the number of family members and toppings. The toppings are numbered $1, 2, \dots, m$.

After this, there are n lines describing the wishes. Each line has two wishes of the form "+ x " (topping x is good) or "- x " (topping x is bad).

Output

Print a line with m symbols: for each topping "+" if it is included and "-" if it is not included. You can print any valid solution.

If there are no valid solutions, print "IMPOSSIBLE".

Constraints

- $1 \leq n, m \leq 10^5$
- $1 \leq x \leq m$

▼ Logical Test Cases

Test Case 1
INPUT (STDIN)
3 5 + 1 + 2 - 1 + 3 + 4 - 2
EXPECTED OUTPUT
- + + + -

Test Case 2
INPUT (STDIN)
3 5 + 2 + 2 - 3 + 3 + 5 - 2
EXPECTED OUTPUT
- + - - +

▼ Mandatory Test Cases

Test Case 1
KEYWORD
void link(int i,int j)

▼ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
1

Test Case 2
TOKEN COUNT
600

Test Case 3
NLOC
71